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Thermodynamic modeling of the ternary Ce–Fe–Sb system: Assessment of the Ce–Sb and Ce–Fe systems

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Abstract

The two Ce–Sb and Ce–Fe binary systems have been evaluated using the calculation of phase diagram method (CALPHAD). All of the binary compounds are treated as stoichiometric compounds. Solution phases are described with an ordinary substitutional solution model. The model parameters were derived from an optimization procedure using all available experimental data. The reproduction of the thermochemical and phase diagram information is reported in a series of figures and tables.
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Keywords: Thermodynamic properties; Ce–Sb system; Ce–Fe system; Phase diagram

1. Introduction

The $CeFe_4Sb_{12}$ ternary intermetallic phase has been evidenced in the ternary system, but no binary ones are present in the borders, as $CoSb_3$ in the Co–Sb system. This intermetallic phase is a new thermoelectric material with the skutterudite structure (space group Im3) [1]. Replacing iron by cobalt or nickel in the compound changes the physical properties of the materials. Cerium–antimony and iron–antimony systems are two edge borders of the cobalt–cerium–iron–antimony multicomponent system. In fact, a good knowledge of the phase relationships as a function of the temperature is very important for understanding the properties and physical behaviour of such materials. The present study was undertaken with that objective.

2. Thermodynamic models

The Gibbs energy of the pure element i, referred to as the enthalpy of its stable state at 298.15 K, is described as a function of temperature by

$$ {}^0G_i - {}^0H_i^{\mathrm{SER}}(298.15\ \mathrm{K}) = a + bT + cT\ln(T) + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9}. \quad (1)$$

The coefficients a through h are taken from the SGTE data set [2].

2.1. Substitutional solutions

The liquid, face-centered cubic (fcc_A1) and body-centered cubic phases (bcc_A2) are described with an ordinary substitutional solution model for which the Gibbs energy expression is given by the following equation:

$$ G_m^{\phi} = x_i\,{}^0G_i^{\phi} + x_j\,{}^0G_j^{\phi} + RT(x_i\ln x_i + x_j\ln x_j) + {}^EG_m^{\phi} \quad (2)$$

where ${}^0G_i^{\phi}$ and ${}^0G_j^{\phi}$ are the molar Gibbs energies of pure substances i and j with ϕ structure. The terms x_i and x_j denote mole fractions of pure substances i and j, respectively. ${}^EG_m^{\phi}$ is the excess Gibbs energy, expressed in Redlich–Kister polynomials [3] as

$$ {}^EG_m^{\phi} = x_i x_j \sum_v {}^vL_{i,j}^{\phi}(x_i - x_j)^v \quad (3)$$

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Table 1
Optimized parameters describing the thermodynamic properties of the Ce–Sb system

Redlich–Kister formalism		
Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ce,Sb}} = -237\,367.5 + 5.0944T$	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ce,Sb}} = -7623.1$
fcc_A1	${}^{0}L^{\mathrm{fcc}}_{\mathrm{Ce,Sb}} = 1000\,000$	
bcc_A2	${}^{0}L^{\mathrm{bcc}}_{\mathrm{Ce,Sb}} = -239\,461.9$	
Stoichiometric phase		
Ce_2Sb	${}^{0}G^{\mathrm{Ce_2Sb}}_{\mathrm{Ce:Sb}} - 2\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} - {}^{0}G^{rho}_{\mathrm{Sb}} = -357\,150.8 + 89.2044T$	
Ce_4Sb_3	${}^{0}G^{\mathrm{Ce_4Sb_3}}_{\mathrm{Ce:Sb}} - 4\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} - 3\,{}^{0}G^{rho}_{\mathrm{Sb}} = -836\,920.7 + 146.0288T$	
CeSb	${}^{0}G^{\mathrm{CeSb}}_{\mathrm{Ce:Sb}} - {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} - {}^{0}G^{rho}_{\mathrm{Sb}} = -237\,661.1 + 31.5989T$	
$CeSb_2$	${}^{0}G^{\mathrm{CeSb_2}}_{\mathrm{Ce:Sb}} - {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} - 2\,{}^{0}G^{rho}_{\mathrm{Sb}} = -313\,562.4 + 70.6549T$	

The values are given in SI units per mole of formula units.

where ${}^{v}L^{\phi}_{i,j}$ is a binary interaction parameter, which will be evaluated in the present work. It is normally described as a linear function of temperature:

$$ {}^{v}L^{\phi}_{i,j} = A_i + B_i T \tag{4} $$

The solution exhibits antiferromagnetic ordering in the fcc_A1 phase and ferromagnetic ordering in the bcc_A2 phase. According to Hillert and Jarl [4], the magnetic contribution to the Gibbs energy is described by

$$ {}^{mg}G^{\phi}_{m} = RT\ln(\beta^{\phi}+1)f(\tau), \quad \tau = T/T^{\phi}_{\mathrm{c}} \tag{5} $$

where T^{ϕ}_{c} is the critical Curie or Neel temperature and β^{ϕ} is the magnetic moment per mole of atoms. The term $f(\tau)$ represents a polynomial obtained by Hillert and Jarl [4]. The detailed expressions for $f(\tau)$ can be found in the literature [4]. In order to describe the composition dependence of the magnetic ordering temperature, the magnetic moment and the ferromagnetic transition temperature were assumed to decrease linearly from pure iron.

2.2. Stoichiometric intermetallic compounds

In this study, the Ce_2Sb, Ce_4Sb_3, CeSb, $CeSb_2$, Ce_2Fe_{17}, and $CeFe_2$ phases are treated as stoichiometric compounds. Their Gibbs energy per mole of formula can be expressed as:

$$ G^{\mathrm{Ce}_A\mathrm{Sb}_B}_{m} = A\,{}^{0}G_i + B\,{}^{0}G_j + a_1 + b_1 T \tag{6} $$

where a_1 and b_1 are lattice stability parameters to be evaluated in the present work.

3. Experimental information of phase diagram and thermodynamics

The Ce–Sb phase diagram given by Massalski [5] was drawn schematically based on a presumed similarity to the Nd–Sb system. Five compounds $CeSb_2$, Ce_5Sb_3, Ce_3Sb_2, Ce_4Sb_3, CeSb, and $CeSb_2$ were reported [6–8]. Abulkhaev [9], using thermal, microscopic, and x-ray methods, studied the Ce–Sb binary phase diagram. The existence of four compounds $CeSb_2$, Ce_4Sb_3, and CeSb, $CeSb_2$ was confirmed. The compounds $CeSb_2$, Ce_4Sb_3, and $CeSb_2$ are formed by peritectic reactions at 1603, 1783 and 1403 K, respectively. There is a polymorphic change for $CeSb_2$ at 1173 K. The compound CeSb melts congruently. The homogeneity ranges of the compounds in the Ce–Sb system are narrower than ±0.5 at.%, and that is why they were treated as line compounds. No evidence of the existence of the Ce_5Sb_3 and Ce_3Sb_2 compounds was found. This system also contains liquid and δCe (bcc_A2), γCe(fcc_A1), and βCe(dhcp), (Sb) terminal solid solutions. Abulkhaev [9] proposed that the solubility of cerium in antimony as well as antimony in cerium are less than 1 at.%. Yamshchikov [10] determined the solubility of cerium in liquid antimony in the temperature range 900–1100 K. The heats of formation in the Ce–Sb system were measured at 300 K by Borsese' group [6,11,12] using a direct isoperibol calorimeter. The following values were found for the various compounds (plus or minus 4 kJ/mol): Ce_2Sb, −102.9 kJ/mol; Ce_5Sb_3, −108.8 kJ/mol; Ce_4Sb_3, −116.7 kJ/mol; CeSb, −126.4 kJ/mol; and $CeSb_2$, −90.0 kJ/mol. The heats of formation of CeSb given by Borsese was confirmed by Schiffman [13] from thermodynamic measurements in the temperature range 1985–2172 K. Experimental data for the heat capacity and entropy of the compounds are lacking. As for the liquid, the only experimental thermodynamic measurements were the properties of the dilute solution reported by Yamshchikov [14,15].

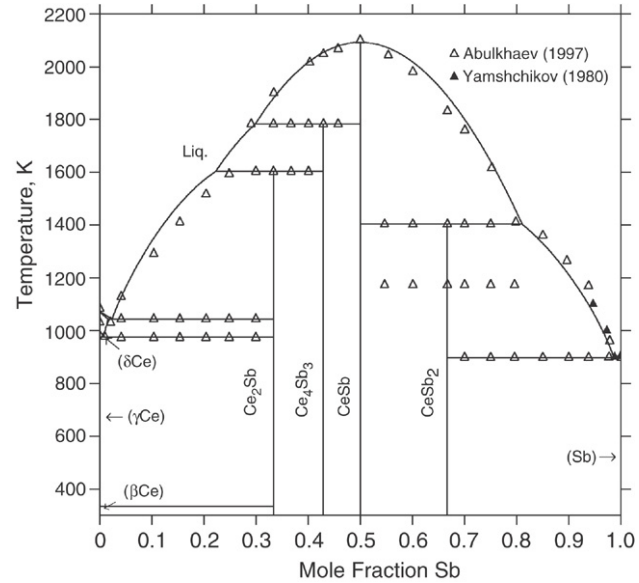

Fig. 1. The Ce–Sb phase diagram calculated from the present thermodynamic description in comparison with experimental measurements.

Vogel [16] indicated the existence of two compounds $CeFe_2$ and Ce_2Fe_5, but many investigators confirmed the existence of two stable phases: cubic $CeFe_2$ with the $MgCu_2$ structure and $CeFe_7$. Gschneidner [17] measured the Ce–Fe binary phase diagram. It is generally agreed that the two stable intermetallic compounds in this system are $CeFe_2$ and Ce_2Fe_{17}. Using thermal, microscopic, and x-ray methods, Chuang [18,19] studied the Ce–Fe binary phase diagram.

Table 2
Invariant reactions in the Ce–Sb system

Reaction	Temperature (K)	Composition, at.% Sb			Reference
$(\delta Ce) \leftrightarrow (\gamma Ce) + Ce_2Sb$	975	0.76	0	33.34	This work
	973			33.34	[9]
$Liq. \leftrightarrow (\delta Ce) + Ce_2Sb$	1044	2.36	1.63	33.34	This work
	1043			33.34	[9]
$Liq. + Ce_4Sb_3 \leftrightarrow Ce_2Sb$	1603	22.24	42.86	33.34	This work
	1603		42.86	33.34	[9]
$Liq. + CeSb \leftrightarrow Ce_4Sb_3$	1784	29.90	50	42.86	This work
	1783		50	42.86	[9]
$Liq. \leftrightarrow CeSb$	2092	50	50		This work
	2093	50	50		[9]
$Liq. + CeSb \leftrightarrow CeSb_2$	1403	81.05	50	66.67	This work
	1403		50	66.67	[9]
$Liq. \leftrightarrow (Sb) + CeSb_2$	897	98.58	100	66.67	This work
	898		100	66.67	[9]

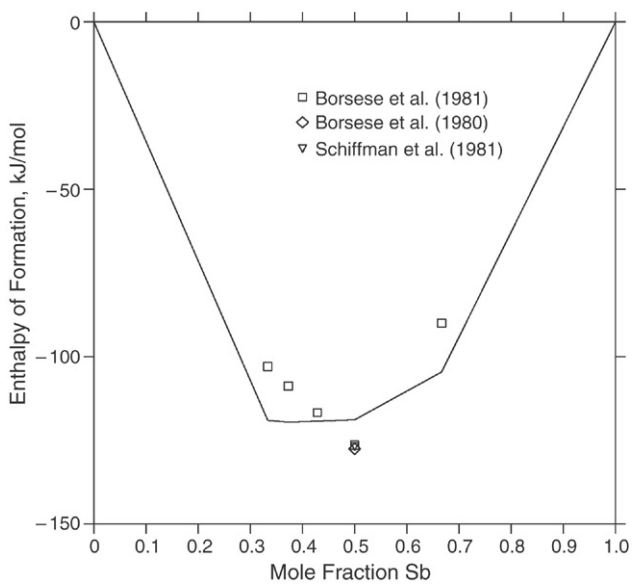

Fig. 2. Calculated enthalpies of formation of the Ce–Sb intermediate phases at 298 K and experimental results.

The phase diagram given by Chuang is similar to the one given by Gschneidner [17]. Consequently, this system contains liquid, peritectic compounds Ce_2Fe_{17} and $CeFe_2$, and terminal solid phases αFe (bcc_A2), δFe (bcc_A2), γFe (fcc_A1), δCe (bcc_A2), γCe (fcc_A1) and βCe (dhcp). The solid solubility of cerium in (αFe) is 0.038 at.% at 1173 K, 0.035 at.% at 1123 K [18,19] and 0.019 at.% at room temperature [20,21]. Li et al. [22] and Yan et al. [23] proposed close values. The values given by Vogel [16] and Jepson [24] are 5.16–6.57 at.% cerium and 0.16 at.% cerium respectively, but these data are not accepted in the assessment. Additions of cerium increase the transformation temperature of δFe $\rightarrow$ γFe from 1667 to 1672 K and that of γFe $\rightarrow$ αFe from 1185 to 1195 K [18, 19]. Ce_2Fe_{17} occurs in two modifications: hexagonal and rhombohedral [17]. Since the transition temperature has not been established, the Ce_2Fe_{17} was treated as one phase. A eutectic reaction occurs at 865 K with the composition 83 at.% cerium [25] or 83.3 at.% cerium [18,19]. Esin [26] determined the mixing enthalpies of liquid in the Ce–Fe system in the composition range 0–35 at.% Ce at 1900 K. The following values were found for the various compositions: 5 at.% Ce, -0.21 ± 0.02 kJ/mol; 10 at.% Ce, -0.42 ± 0.04 kJ/mol; 15 at.% Ce, -0.61 ± 0.05 kJ/mol; 20 at.% Ce, -0.69 ± 0.06 kJ/mol; 25 at.% Ce, -0.55 ± 0.07 kJ/mol; 30 at.% Ce, -0.06 ± 0.07 kJ/mol; and 35 at.% Ce, 0.74 ± 0.07 kJ/mol. Burylev [27] measured the activities of this system at 1500 K. No experimental measurements for thermodynamic properties of the compounds in the Ce–Fe system have been reported.

Table 3
Optimized parameters describing the thermodynamic properties of the Ce–Fe system

Redlich–Kister formalism		
Liquid	${}^0L^{Liq}_{Ce,Fe} = 5881.8 - 8.5586T$	${}^1L^{Liq}_{Ce,Fe} = 12585.0 - 20.1552T$
	${}^2L^{Liq}_{Ce,Fe} = 2591.0$	
fcc_A1	${}^0L^{fcc}_{Ce,Fe} = 59316.5$	
bcc_A2	${}^0L^{bcc}_{Ce,Fe} = 3736.6 + 35.2022T$	
Stoichiometric phase		
Ce_2Fe_{17}	${}^0G^{Ce_2Fe_{17}}_{Ce:Fe} - 2\,{}^0G^{fcc}_{Ce} - 17\,{}^0G^{bcc}_{Fe} = -184\,007.7 + 107.7769T$	
$CeFe_2$	${}^0G^{CeFe_2}_{Ce:Fe} - {}^0G^{fcc}_{Ce} - 2\,{}^0G^{bcc}_{Fe} = -26\,825.0 + 9.8533T$	

The values are given in SI units per mole of formula units. The reference state for Fe is pBcc_A2 (the paramagnetic state).

4. Assessment procedure

As the first step for the Ce–Sb system, the thermodynamic parameters for the liquid and congruent CeSb compound were optimized based on the data of the liquidus and the formation enthalpy of CeSb compound using the Thermo–Calc program [28]. For the liquid, it was found that three coefficients,

Table 4
Invariant reactions in the Ce–Fe system

Reaction	Temperature (K)	Composition, at.% Fe			Reference
Liq. + (δFe) ↔ (γFe)	1672	90.58	99.89	99.88	This work
	1672	88.3	~1	~1	[18,19]
	1673	88.7			[17]
(γFe) + Ce_2Fe_{17} ↔ (αFe)	1192	99.93	89.47	99.98	This work
	1195	~1	89.47	~1	[18,19]
	1198				[17]
Liq. + (γFe) ↔ Ce_2Fe_{17}	1336	58.96	99.90	89.47	This work
	1336	59.5	~1	89.47	[18,19]
	1341	57.6	~1	89.47	[17]
Liq. + Ce_2Fe_{17} ↔ $CeFe_2$	1198	46.36	89.47	66.67	This work
	1198	45.9	89.47	66.67	[18,19]
	1205	45.0	89.47	66.67	[17]
Liq. ↔ Ce_2Fe_{17} + (γCe)	865	16.35	89.47	0.01	This work
	865	16.7	89.47	~0	[18,19]
	865	17.0	89.47	~0	[17]
Liq. + (δCe) ↔ (γCe)	999	4.20	0.03	0.01	This work
	998	8.7	~0	~0	[17]

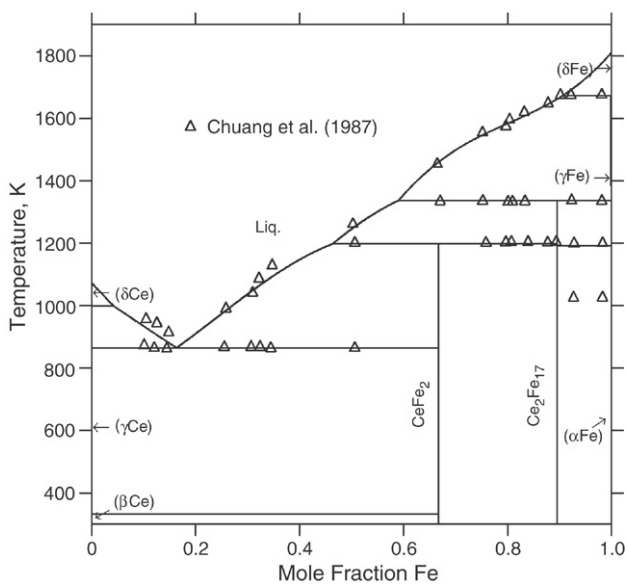

Fig. 3. The Ce–Fe phase diagram calculated from the present thermodynamic description compared with experimental measurements.

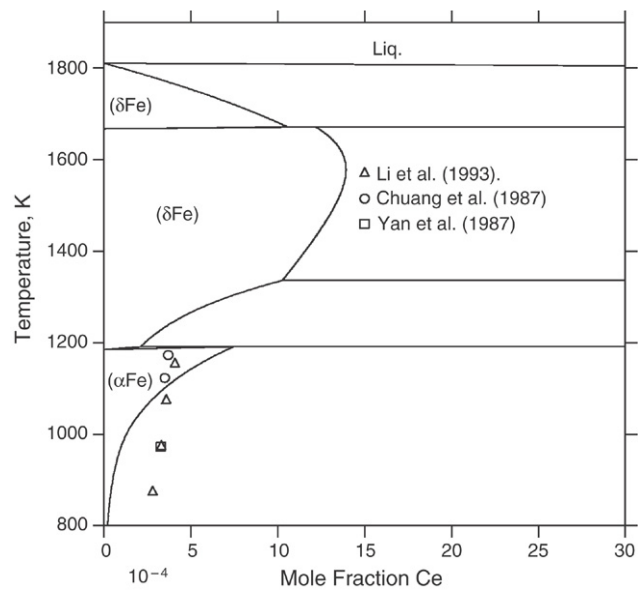

Fig. 4. The assessed solvus of cerium in iron compared with experimental measurements.

A_0, B_0 and A_1, were introduced into Eq. (4) because no thermochemical data for the liquid phase are known at present. In the case of Ce_2Sb, Ce_4Sb_3, CeSb and $CeSb_2$, it is possible to optimize independent coefficients a_1 and b_2 in Eq. (5) because both the enthalpy of formation and phase diagram data are available. For the bcc_A2 phase, coefficient A_0 alone in Eq. (4) is enough. For fcc_A1, no solubility is assumed, which is realized by employing a large positive interaction parameter, i.e. ${}^0L^{\mathrm{fcc}}_{\mathrm{Ce,Sb}} = 1000\,000\ \mathrm{J\ mol^{-1}}$. During the course of the optimization, the weights were adjusted systematically to improve the overall consistency of the data.

All the parameters were finally evaluated together and adjustments were made to give the best description of the system. A similar procedure was taken for the Ce–Fe system.

5. Results and discussion

5.1. Ce–Sb system

All the parameters for the Ce–Sb system obtained by means of the optimization procedure are listed in Table 1. The refined phase diagram for the binary Ce–Sb system following the assessment is shown in Fig. 1 with the available experimental data superimposed. It can be seen that the calculated phase boundaries are in good agreement for all the experimental data [9].

As can be seen in Table 2, the temperatures and compositions for all the invariant equilibria in the system are well reproduced. The assessed liquidus agrees reasonably well with the results of Abulkhaev [9].

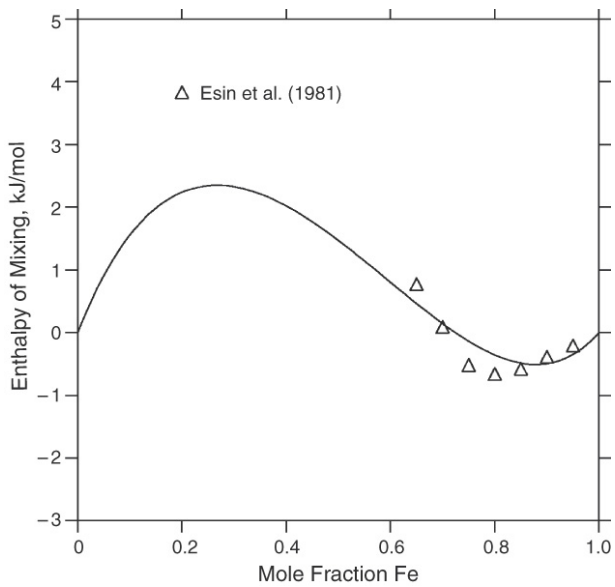

Fig. 5. Calculated mixing enthalpy of liquid in the Ce–Fe at 1900 K.

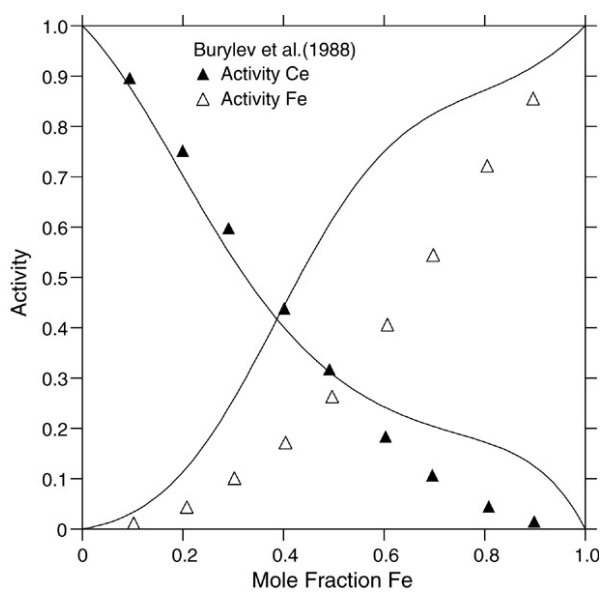

Fig. 6. The calculated activities at 1500 K with experimental results.

The assessed enthalpies of formation are plotted in Fig. 2. The experiment data from references [6,11–13] are also shown in Fig. 2. As can be seen, the calculated enthalpies of formation of Ce_2Sb, Ce_4Sb_3, CeSb and $CeSb_2$ agree reasonably well with the experimental data.

5.2. Ce–Fe system

All the evaluated parameters of the Ce–Fe system are listed in Table 3. The assessed phase diagram is shown in Fig. 3 with experimental values from reference [18,19]. The calculated phase diagram is in good agreement with the experiments. All invariant equilibria in the system are listed in Table 4. The recommended data by Gschneidner [17] and Chuang [18,19] are also quoted in Table 4. The values of the calculated invariant equilibrium temperatures are almost

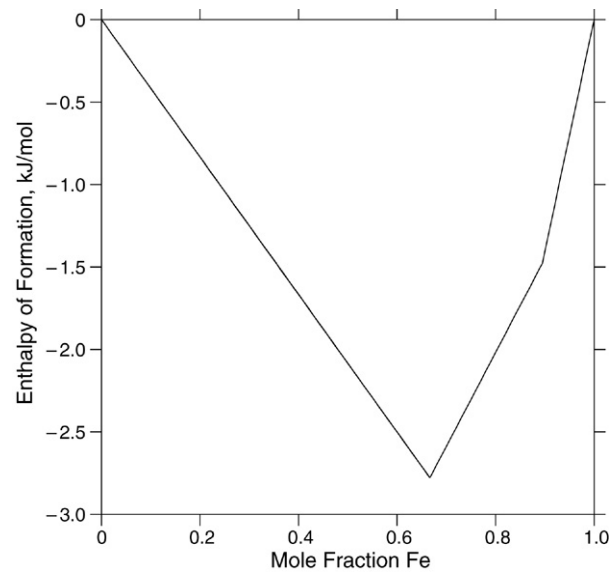

Fig. 7. Calculated enthalpies of formation of the Ce–Fe intermediate phases at 298 K.

the same as the corresponding experimental data [17–19]. As can be seen in Fig. 4, the assessed solvus of cerium in iron agrees with the experimental data [18,19,22,23]. Fig. 5 shows the calculated mixing enthalpies of liquid in the Ce–Fe system at 1900 K. The agreement with the experimental measurements is satisfactory [26]. Burylev [27] reported a much smaller activity of iron in Fig. 6. During the course of the optimization, attempts were made to fit the data from Burylev [27], but it made the reproduction of the phase diagram impossible. These attempts revealed that the data are incompatible with other thermodynamic and phase diagram measurements. The excess Gibbs energy of the Ce–Fe liquid is uncertain due to the uncertainty in the enthalpy and activity measurements. Fig. 7 shows the calculated enthalpies of formation at 298 K. Nikolaenko [29] indicated that gradual growth of the TM–R interaction along the R series was observed. The minima of molar enthalpy of mixing for the Fe–Gd system is −7.4 kJ/mol [29]. The formation enthalpy of Pr_2Fe_{17} measured by Stefan et al. [30] is about −2 kJ/mol. Therefore, the calculated enthalpy of formation and enthalpy of mixing are reasonable.

6. Conclusions

The phase relations and thermodynamic properties of the Ce–Sb and Ce–Fe systems were evaluated from the experimental information available in the literature. Consistent sets of thermodynamic parameters have been derived and all the experimental data have been reproduced reasonably. But, the experimental data of the entropy and the heat capacity for the compounds in the systems are lacking. The estimation of the Gibbs energy of the liquid in the Ce–Sb system was not based on any experimental measurements. More experimental work on the thermodynamic properties of the systems may be necessary to improve the description.

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